Makkapati Meghana

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.81
Sri Vasishta Educational Institutions	Class XII (Telangana State Board)	2023	9.78%
Matrusri DAV Public School	Class X (CBSE)	2021	9.60%

WORK EXPERIENCE

WizLearnr International Pvt. Ltd. -- GenAI / EdTech Startup

Technical Intern, AI

- Worked on **AI/ML systems spanning computer vision, LLM compression, and multimodal model optimization**, taking projects from experimentation and training through evaluation and deployment.
- Built and evaluated models under **accuracy, latency, memory, and energy constraints**, using quantitative benchmarking and **vLLM-based inference** to optimize model deployment.
- Developed end-to-end ML pipelines involving **transfer learning, knowledge distillation, LoRA/QLoRA, quantization, and inference optimization**.
- Conducted systematic evaluation across multiple task categories and analyzed failure cases to improve **model performance, inference efficiency, and deployment readiness**.

PROJECTS

On-device sketch recognition with MobileNetV3

- Built an on-device sketch classification pipeline for a children's vocabulary app, scaling the dataset from **100 to 400 classes** under strict size, latency, and power constraints.
- Fine-tuned **ImageNet-pretrained MobileNetV3** using transfer learning and a 3-phase gradual unfreezing strategy, improving held-out accuracy from **94% to 95.02%** *(82% Top-1 / 97% Top-5 at 100 classes)*.
- Reduced overfitting using **early stopping, dropout, label smoothing, and targeted augmentation**, and benchmarked **MobileNetV3 vs. EfficientNet-Lite0** on accuracy, latency, and training stability.

Quantizing Sarvam 30B *(UNESCO Resilient AI Challenge 2026)*

- Compressed a **120 GB, 30B-parameter multilingual MoE LLM to 19 GB (6.3× smaller)** while retaining **98.3% of FP16 judged quality**, exceeding the challenge's 80% quality floor.
- Built an end-to-end **distillation and quantization pipeline** using self-distillation, **LoRA/QLoRA**, and **GPTQ W4A16** with an independent 1,200-prompt calibration set.
- Reduced inference energy by **33.3%** and latency by **1.4×**, increasing throughput from **175 to 253.9 tokens/s** on an **A100 80GB with vLLM**; evaluated performance across six task buckets.

Foundation Model for Cell Instance Segmentation

- Developing a **foundation model for cell instance segmentation and counting**, designed to separate densely packed and touching HeLa cells from limited labelled data (using 27 labelled frames)
- Implemented architectural changes and finetuning techniques on U-Net and HRNet-Lite and benchmarked different models under a common 5-fold evaluation framework.
- Achieved **~0.70 AP@0.5** with Attention U-Net while prioritizing **low inference latency and compact model size** for efficient deployment.

HACKATHONS

UNESCO Resilient AI Challenge 2026

- Independently developed an LLM compression pipeline that reduced Sarvam 30B from **120 GB to 19 GB (6.3×)** while retaining **98.3% of FP16 judged quality**.

POSITIONS OF RESPONSIBILITY

Coordinator, Corporate Partnerships and Growth Domain, Finance and Consulting Club (2024-2025)

Oversee the development and management of corporate engagements to secure financial and strategic support for the club's initiatives.

EXTRACURRICULAR

Basketball NSO Team Member, IIT Hyderabad (2023-Present): Trained and competed in institute-level basketball under professional coaching.

Member, Vibes - The Music Club, IIT Hyderabad (2023-2024): Performed and organized musical events, promoting a vibrant campus music culture.

SKILLS

Languages: Python, C, C++, JavaScript, HTML / CSS, SQL.

ML / DL: TensorFlow / Keras, scikit-learn, OpenCV, Hugging Face Transformers, LoRA, GPTQ, vLLM; transfer learning, knowledge distillation, quantization (QLoRA, GPTQ W4A16), quantization-aware training, hyperparameter tuning, model benchmarking.

Tools: Git / GitHub, Pandas, NumPy, Matplotlib, Seaborn, Plotly, CodeCarbon, Vertex AI